SkyHAD: Cooperative Aerial Asset Defense and Sixteen Native Task Formulations

***Abstract*—Cooperative Red defenders protect stationary assets against Blue opponents in a three-dimensional scene. Eight native model–action interfaces and two objectives define sixteen tasks. Particle, fixed-wing and quadrotor dynamics give different control burdens to acceleration, position and actuator commands. Survival protects every asset through a terminal team payoff; Damage minimizes cumulative raw asset damage. This formulation distinguishes local actor information, privileged critic features, complete transition state and external opponent policies, and states the assumptions for decentralized execution and centralized training.**

***Index Terms*—Multi-agent reinforcement learning, cooperative defense, decentralized control, aerial simulation.**

I. Scenario and Research Goal

Red agents coordinate interception and protection of stationary assets. Blue agents navigate under a specified external opposing policy; automatic Attack fire and other role interactions are resolved independently of navigation commands. The ultimate research goal is to learn effective cooperative Red defense with decentralized policies acting on local information. Training may use explicitly provided centralized features.

Three physical models are available: a six-value kinematic particle, a six-degree-of-freedom aerodynamic fixed wing, and a six-degree-of-freedom ideal quadrotor. Model type differs from the three episode-fixed functional roles: Attack, Scout and Disturb. Every role uses its selected model and navigation interface; functional decisions follow shared environment rules.

The particle supports acceleration and absolute-position actions. Each rigid model supports acceleration, direct actuator and absolute-position actions. These are eight native model–action combinations. Pairing each with Survival and Damage gives the sixteen themes indexed on page 4.

A. Task distinction

Survival uses default terminal Red payoff ± 10 with asset-loss priority. Damage accumulates raw asset losses while finite asset health remains unchanged. Red extinction alone is not a native global terminal event. The sample horizon H limits collected trajectories; it does not replace the continuing objective by a finite-horizon game.

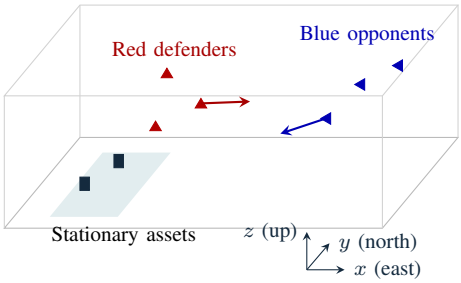


Fig. 1. Illustrative 3D asset-defense scene. Arrows show navigation directions; no sensing or firing range is depicted.

B. Scope and reading order

Pages 2–4 define shared dynamics, ordered events, information, tasks and learning. Each of the sixteen themes is one continuous two-page section: its controlled physical model and reset lead to defender information, a frozen-opponent process, task payoff and policy optimization. Only particle themes admit planar compatibility; the native formulations are 3D. The document has 36 pages.

The implementation grounds this specification: particle position advances with old velocity; all living entities appear in native actor rows; automatic firing is separate from external Blue navigation; and fixing Blue is an explicit experiment choice. Physical fidelity and action abstraction are experimental choices, so the control burden differs across interfaces.

C. Units and model assumptions

Distances are metres, speeds m/s, rates rad/s and time seconds. λ scales scene geometry and role radii, never aircraft mass/inertia. $\mathrm{sat}_L(z)$ limits Euclidean norm; clip limits components/scalars. Each theme supplies its own ρ_0, physical parameters, controller and action interface.

D. Implementation correspondence and assumptions

The current flat implementation grounds this specification: `config.py` fixes scales and limits; `initialization.py` defines reset sampling; `dynamics/` defines particle/rigid maps, controllers and physical constants; `world.py` and `simulation.py` resolve events and motion; `observations.py` packs information; `tasks.py` defines rewards and completion; `grouping/` defines the optional extension. `docs/MODELING.md` records the same modeling assumptions.

Physical parameters and source-derived attitude gains are attributed in the physics sections [1], [2], [3]. Interface guidance, action scalings, finite asset health, boundaries and task objectives are project choices. No wind, observation noise, actuator lag or unlisted vehicle drag is assumed. These sources establish model assumptions, not training results for the sixteen instances.

II. Common transition and physical state

A. Sets, symbols and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N+K$. Role $r_i \in \{A, S, D\}$ (Attack, Scout, Disturb), side $c_i \in \{R, B\}$ and roster identity id_i are episode-fixed. Set subscripts select roles.

$$\begin{aligned} s_t = \big(&\{x_i, h_i, r_i, c_i, \mathrm{id}_i\}_{i\in\mathcal{U}}, \\ &\{p_k, h_k, D_k\}_{k\in\mathcal{P}}, t, \zeta_t\big). \end{aligned} \tag{1}$$

Particle $x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$; rigid $x_i = (p_i, v_i, q_i, \omega_i) \in \mathbb{R}^{13}$, $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

B. Six-degree-of-freedom dynamics

Use world ENU and body FLU axes. The scalar-first unit quaternion $q = (q_w, q_x, q_y, q_z) \sim -q$ maps body to world through $R(q) \in SO(3)$. The 13 stored values represent a six-DOF rigid body with a 12-dimensional state: six configuration and six velocity coordinates.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \tfrac{1}{2} q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned} \tag{2}$$

Here $g = 9.81$ m/s^2, ω is body angular velocity (rad/s), and F_B excludes gravity; wind, sensor noise and actuator lag are absent.

$$\begin{aligned} R(q) &= I + 2q_w[q_v]_\times + 2[q_v]_\times^2, \\ q_v &= (q_x, q_y, q_z). \end{aligned} \tag{3}$$

Here $[z]_\times y = z \times y$.

C. Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q. An external step is $\Delta = 1$ s, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned} \tag{4}$$

Repeat n times with normalized stage quaternions. Hold actuators or intent references; recompute intent feedback at each stage. Controllers have no hidden integrators.

Clip endpoint $p'_a = \mathrm{clip}(p^*_a, b^-_a, b^+_a)$. At $p^*_a > b^+_a$ set $v'_a = \min(v^*_a, 0)$; at $p^*_a < b^-_a$ set $v'_a = \max(v^*_a, 0)$; otherwise retain v^*_a. Clip collision-path substep positions too. Controller reference limits do not clamp physical airspeed or attitude.

D. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau\Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\begin{aligned} \tau_* &= \begin{cases} 0 & \|d\|^2 \le 10^{-12}, \\ \mathrm{clip}(-r^T d/\|d\|^2, 0, 1) & \text{otherwise}, \end{cases} \\ d_* &= \|r + \tau_* d\|. \end{aligned} \tag{5}$$

Collision sets both UAV HP to zero when $\min d_* \le d_c$; targets are excluded. Pairs use i-major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep.

After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k\in\mathcal{P}: h_k>0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j\in\mathcal{B}_A: h_j>0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases} \tag{6}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \le d < r_1, \\ 0 & d \ge r_1, \end{cases} \tag{7}$$

$$\begin{aligned} L_k &= \sum_{b\in\mathcal{B}_A: h_b>0} I_b \eta(\|p_b - p_k\|), \\ L_i^{\mathrm{atk}} &= \sum_{j\in\mathcal{U}_A: c_j\neq c_i, h_j>0} I_j \eta(\|p_j - p_i\|). \end{aligned} \tag{8}$$

A source can affect multiple enemies, excluding friendly explosions. Sum losses on one post-collision, pre-motion snapshot before clipping HP.

E. Disturb and Scout

For source j, $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3}. Let

$$\begin{aligned} G_{ji} &= \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ &\quad \times e^{-(d_{ji}/600\lambda)^2/2}/\sqrt{2\pi}, \\ S_c(j) &= \sum_{i\in\mathcal{U}: c_i=c, h_i>0} G_{ji}. \end{aligned} \tag{9}$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \ge S_B(j)]$. Dead sources contribute zero.

$$\begin{aligned} L_i^{\mathrm{dis}} &= .1 \sum_{j\in\mathcal{U}_D: h_j>0} J_j G_{ji}, \\ \widetilde{h}_i &= \max(0, h_i - L_i^{\mathrm{atk}} - L_i^{\mathrm{dis}}). \end{aligned} \tag{10}$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \widetilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step. The native order is collision, role decision, losses and source consumption, motion, asset/task update, cycle increment, then reward, completion/truncation and observation. Snapshot pre-step participation; retain commands, flags and raw damage for replay.

III. Common information and defender task

A. Observation and information structure

For each living entity $j \neq i$, the actor row is

$$\begin{aligned} e_{ij} = \big[&(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ &h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j\in\mathcal{P}}\big], \\ L = &\sqrt{\sum_a (b_a^+ - b_a^-)^2}. \end{aligned} \tag{11}$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

Interface-specific self blocks and dimensions are stated in each theme. With deterministic encoding, $Z_i(o \mid s) = \delta_{\mathrm{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) =$ `state()` packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P], \tag{12}$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\mathrm{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

B. Fixed-opponent defender Dec-POMDP

For specified frozen external navigation policy π_B, including its memory transition, define

$$\begin{aligned} \mathcal{M}_R &= \big(\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H\big), \\ T^{\pi_B}(s' \mid s, a_R) &= \int \delta_{F_C(s,a_R,a_B)}(s')\pi_B(da_B \mid o_B, \zeta_t). \end{aligned} \tag{13}$$

F_C includes the ordered events, model-specific controlled motion and any memory update. Native execution accepts both teams' navigation actions. Automatic fire is an environment rule and does not supply Blue navigation. A rule-based or learned frozen Blue policy is an explicit experiment choice; with both teams learning the system is a partially observable stochastic game (POSG). The reset law ρ_0 is specified with the physical model in each theme.

C. Configuration and replay state

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

D. Interpretation of partial information

Actor inputs omit other UAV attitudes/rates, policy memory and time. Fixed roster positions and side flags do not encode role directly. Full entity inclusion therefore does not guarantee Markov observations. A recurrent policy's local history can retain its own inferred information; that policy memory also affects exact centralized value targets. Privileged $g(s)$ is an encoding for CTDE, not a complete simulation snapshot or proof of the Markov property.

E. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases} \tag{14}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1-d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 4.

F. Damage payoff and completion

Assets keep finite reset health $h'_k = h^0_k$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t. \tag{15}$$

Blue reward is d_t; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \tag{16}$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t, including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

G. Three distinct masks

Entity validity masks identify living packed rows. Pre-step participation masks decide which actor actions contribute to the action loss, including a final lethal action. The global bootstrap mask suppresses successor value only at natural world completion. Actor retirement, entity death, sample truncation and global completion are separate events. When native completion coincides with H, natural completion takes priority.

IV. Common learning objective and implementation index

A. MARL / CTDE objective and value history

$$\begin{aligned} \tau_i &= (o_{i,0}, a_{i,0}, \ldots, o_{i,t}), \\ \pi_R(a_R \mid \tau_R) &= \prod_{i \in \mathcal{R}} \pi_i(a_i \mid \tau_i), \\ \pi_R^* &\in \arg\max_{\pi_R} \mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t R_t\right]. \end{aligned} \tag{17}$$

Expectation is over ρ_0, T^{π_B}, π_R. Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, the native task is terminal Survival payoff or negative total asset damage; $\gamma < 1$ selects a discounted task. Actors use local histories only. Let ξ_t augment $g(s_t)$ with task state and all relevant policy history/memory, including recurrent Red actor memory and frozen Blue memory. Exact value targets condition on ξ_t; $g(s)$ alone need not suffice:

$$y_t = R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \tag{18}$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or explicit memory/state augmentation supplies the required conditioning; privileged packed features do not automatically define a Markov critic state. This defines an objective; it does not prescribe an implemented optimizer.

B. Optional grouping / SMDP extension

Grouping adds assignments, reserves, decision clocks and policy memory to state, and currently executes acceleration commands. It does not add native themes. Macro duration d ends at a command opportunity, casualty or completion. With no living Red, the call continues Blue scheduling to completion/horizon and records `no_red_continuation=True`.

$$\bar{R}_t = \sum_{j=0}^{d-1} \gamma^j R_{t+j}, \qquad y_t = \bar{R}_t + \gamma^d b_t V(\xi_{t+d}). \tag{19}$$

Current grouping Damage returns the *undiscounted* sum, directly matching $\gamma = 1$. A γ^d bootstrap alone does not discount within-macro rewards. Grouping Survival uses success 0/1 and a configured horizon result (`red_win`, `draw`, `blue_win`); native Survival instead uses ± 10 and sampling truncation. Grouping Damage retains sampling truncation.

Built-in fixed Blue opponents are reactive, balanced and concentrated with a rush executor. Balanced uses target quotas; concentrated selects minimum mean-travel target. Reactive target probability is proportional to $e^{-2\,\mathrm{coverage} - .25\,\mathrm{travel}}$, with coverage $\sum_{i \in \mathcal{R}: h_i > 0} h_i e^{-d_{ik}/(850\lambda)}$ and travel equal to mean Blue distance divided by 5000λ. Native seedless resets advance the local RNG; grouping seedless reset reproduces its saved seeded initialization.

TABLE I
Eight native model–action combinations: theme IDs and generated ranges of the sixteen continuous two-page sections.

Model	Action	Survival	Damage
Particle	Acceleration	1: 5–6	2: 7–8
Particle	Position	3: 9–10	4: 11–12
Fixed wing	Acceleration	5: 13–14	6: 15–16
Fixed wing	Actuator	7: 17–18	8: 19–20
Fixed wing	Position	9: 21–22	10: 23–24
Quadrotor	Acceleration	11: 25–26	12: 27–28
Quadrotor	Actuator	13: 29–30	14: 31–32
Quadrotor	Position	15: 33–34	16: 35–36

C. Optional native Survival rewards

Each formulation uses its native default payoff. An explicit four-vector w replaces Survival scalar reward by $w^T \ell_i$, where $\ell_i = (r_i^{hit}, r_i^{target}, r_i^{enemy}, 10\epsilon_i z)$, where $\epsilon_i = +1$ for Red and -1 for Blue. All geometry below is post-step. For a living Attack agent,

$$\begin{aligned} r_i^{hit} &= \sum_{j \in \mathcal{U}: c_j \neq c_i, h_j > 0} \max(0, 1 - d_{ij}/r_1), \\ r_i^{target} &= -\min_{k: h_k > 0} d_{ik}/L, \\ r_i^{enemy} &= \begin{cases} -\min_{j \in \mathcal{B}: h_j > 0} d_{ij}/L & i \in \mathcal{R}, \\ \min_{j \in \mathcal{R}: h_j > 0} d_{ij}/L - 1 & i \in \mathcal{B}. \end{cases} \end{aligned} \tag{20}$$

An empty comparison set gives zero, including the Blue enemy slot; dead agents have zero first three slots. "Hit" is a geometric proxy, not actual fire damage. Scout/Disturb keep their role scalar in slot 1 and zero slots 2–3. Let δ_i be nearest XY-wall distance and $B_i = .05 e^{-\delta_i/(100\lambda)} \mathbf{1}[\delta_i < 100\lambda]$. Scout slot 1 is $c_s|\mathrm{sector}_i| - B_i$; Disturb slot 1 is $\mathbf{1}[J_i, h_i > 0](.1 c_d \sum_{j \in \mathcal{U}} \mathrm{sign}_{ij} G_{ij} - B_i)$, with enemy sign $+1$, friendly sign -1. Defaults $c_s = c_d = 0$. The episode slot already contains the team payoff; adding it again doubles the terminal reward. Damage rejects these weights.

References

[1] BYU MAGICC, "Published Aerosonde parameter set." [Online]. Available: https://github.com/byu-magicc/mavsim_public/blob/main/mavsim_python/parameters/aerosonde_parameters.py. Physical data only; project force/controller code is independently authored.

[2] T. Lee, M. Leok, and N. H. McClamroch, "Control of Complex Maneuvers for a Quadrotor UAV using Geometric Methods on SE(3)," v2, 2010, Secs. II and VII. [Online]. Available: https://arxiv.org/pdf/1003.2005v2. Axes are rotated to ENU/FLU.

[3] M. Faessler, A. Franchi, and D. Scaramuzza, "Differential Flatness of Quadrotor Dynamics Subject to Rotor Drag for Accurate Tracking of High-Speed Trajectories," RAL, 2018. [Online]. Available: https://rpg.ifi.uzh.ch/docs/RAL18_Faessler.pdf. The ideal project model omits those drag terms.

V. Theme 1 / Particle / Acceleration / Survival

For the Survival defense task, the particle navigation map determines the interception geometry. We first specify that map, its action interface and reset law, then formulate the defender information and payoff under a frozen Blue policy.

A. Physical model and controlled transition

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory.

B. Kinematic navigation map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, \quad p_i' = \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, \quad \widetilde{v}_i = \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \tag{21}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \operatorname{clip}(\|w\|, m, V). \tag{22}$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \widetilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\widetilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update. This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \tag{23}$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 40 a_i$ m/s^2. Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}\,40$; discrete nonzero intents have norm 40. 40 is a control scale, not a clamp on actual rigid-body acceleration. The chosen vector directly enters the particle map as $\alpha_i = 40 a_i$. No inner rigid-body controller or actuator state is introduced. Zero input preserves unconstrained velocity; it does not command a stop.

D. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \tag{24}$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; draws outside $\mathcal{W}$ are rejected and resampled. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Openings within $\max(30\lambda, d_c + 10^{-6})$ of an already placed entity are resampled. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$. This reset law and navigation map are common to Survival and Damage; their distinction enters through asset health, completion and the defender reward below. The information structure therefore reflects the chosen action interface rather than the task label.

E. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. The heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor input is only an $(M-1)\times 11$ entity table ($V_{\max}=120$); it has no own-state block. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\dim o_i = 11(M-1), \qquad \dim g = 11M. \tag{25}$$

The acceleration actor has no self-state block. Relative entity rows alone do not supply its absolute position or velocity, so its policy must infer any needed own-motion information from local history.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For the particle, collision prediction follows old-velocity line segments. Consequently, the selected acceleration changes subsequent velocity and interception geometry after the current position advance. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

H. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10 z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{26}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{27}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

VI. Theme 2 / Particle / Acceleration / Damage

For the Damage defense task, the particle navigation map determines the interception geometry. We first specify that map, its action interface and reset law, then formulate the defender information and payoff under a frozen Blue policy.

A. Physical model and controlled transition

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory.

B. Kinematic navigation map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, \quad p_i' = \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, \quad \widetilde{v}_i = \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \tag{28}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \operatorname{clip}(\|w\|, m, V). \tag{29}$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \widetilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\widetilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update. This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \tag{30}$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 40 a_i$ m/s^2. Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}\,40$; discrete nonzero intents have norm 40. 40 is a control scale, not a clamp on actual rigid-body acceleration. The chosen vector directly enters the particle map as $\alpha_i = 40 a_i$. No inner rigid-body controller or actuator state is introduced. Zero input preserves unconstrained velocity; it does not command a stop.

D. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \tag{31}$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; draws outside $\mathcal{W}$ are rejected and resampled. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Openings within $\max(30\lambda, d_c + 10^{-6})$ of an already placed entity are resampled. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$. This reset law and navigation map are common to Survival and Damage; their distinction enters through asset health, completion and the defender reward below. The information structure therefore reflects the chosen action interface rather than the task label.

E. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. The heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor input is only an $(M-1)\times 11$ entity table ($V_{\max}=120$); it has no own-state block. In Damage, an already-dead observer has a zero complete table. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\dim o_i = 11(M-1), \qquad \dim g = 11M. \tag{32}$$

The acceleration actor has no self-state block. Relative entity rows alone do not supply its absolute position or velocity, so its policy must infer any needed own-motion information from local history.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For the particle, collision prediction follows old-velocity line segments. Consequently, the selected acceleration changes subsequent velocity and interception geometry after the current position advance. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

H. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned} h'_k &= h^0_k, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d^{\mathrm{term}}_t &= \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \le 0]. \end{aligned} \tag{33}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma = 1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d^{\mathrm{term}}_t, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{34}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

VII. Theme 3 / Particle / Position / Survival

For the Survival defense task, the particle navigation map determines the interception geometry. We first specify that map, its action interface and reset law, then formulate the defender information and payoff under a frozen Blue policy.

A. Physical model and controlled transition

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory.

B. Kinematic navigation map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, \quad p_i' = \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, \quad \widetilde{v}_i = \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \tag{35}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \operatorname{clip}(\|w\|, m, V). \tag{36}$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \widetilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\widetilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update. This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Absolute position intent

Each agent selects $a_i = p_i^{\mathrm{des}} \in \mathcal{W}$ in metres. The stateless outer loop computes

$$\begin{aligned} d_i &= p_i^{\mathrm{des}} - p_i, \\ v_i^{\mathrm{des}} &= d_i \min\{.5, 120/\max(\|d_i\|, 10^{-12})\}, \\ \alpha_i &= \mathrm{sat}_{40}(2(v_i^{\mathrm{des}} - v_i)), \\ \mathrm{sat}_L(z) &= z \min\{1, L/\|z\|\}, \quad \mathrm{sat}_L(0) = 0. \end{aligned} \tag{37}$$

The command is an absolute point, not a displacement or teleport. The velocity reference can approach zero, but the particle speed map enforces its nonzero free-flight minimum; this interface does not promise arrival with rest. There are three continuous action coordinates and no position-action quantization.

D. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \tag{38}$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; draws outside $\mathcal{W}$ are rejected and resampled. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Openings within $\max(30\lambda, d_c + 10^{-6})$ of an already placed entity are resampled. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$. This reset law and navigation map are common to Survival and Damage; their distinction enters through asset health, completion and the defender reward below. The information structure therefore reflects the chosen action interface rather than the task label.

E. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. The heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=120$) and the unnormalized six-value self block $[p_i, v_i]$. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\dim o_i = 11(M-1)+6, \qquad \dim g = 11M. \tag{39}$$

The position actor's six raw self values supply the position and velocity required by its absolute-point interface. This information does not remove the minimum free-flight speed or alter the collision transition. The dimensions describe the native 3D particle; planar compatibility is confined to the setting stated above.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For the particle, collision prediction follows old-velocity line segments. Consequently, the selected acceleration changes subsequent velocity and interception geometry after the current position advance. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

H. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), \qquad & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{40}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{41}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

VIII. Theme 4 / Particle / Position / Damage

For the Damage defense task, the particle navigation map determines the interception geometry. We first specify that map, its action interface and reset law, then formulate the defender information and payoff under a frozen Blue policy.

A. Physical model and controlled transition

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory.

B. Kinematic navigation map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, \quad p_i' = \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, \quad \widetilde{v}_i = \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \tag{42}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \operatorname{clip}(\|w\|, m, V). \tag{43}$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \widetilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\widetilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update. This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Absolute position intent

Each agent selects $a_i = p_i^{\mathrm{des}} \in \mathcal{W}$ in metres. The stateless outer loop computes

$$\begin{aligned} d_i &= p_i^{\mathrm{des}} - p_i, \\ v_i^{\mathrm{des}} &= d_i \min\{.5, 120/\max(\|d_i\|, 10^{-12})\}, \\ \alpha_i &= \mathrm{sat}_{40}(2(v_i^{\mathrm{des}} - v_i)), \\ \mathrm{sat}_L(z) &= z \min\{1, L/\|z\|\}, \quad \mathrm{sat}_L(0) = 0. \end{aligned} \tag{44}$$

The command is an absolute point, not a displacement or teleport. The velocity reference can approach zero, but the particle speed map enforces its nonzero free-flight minimum; this interface does not promise arrival with rest. There are three continuous action coordinates and no position-action quantization.

D. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \tag{45}$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; draws outside $\mathcal{W}$ are rejected and resampled. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Openings within $\max(30\lambda, d_c + 10^{-6})$ of an already placed entity are resampled. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$. This reset law and navigation map are common to Survival and Damage; their distinction enters through asset health, completion and the defender reward below. The information structure therefore reflects the chosen action interface rather than the task label.

E. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. The heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=120$) and unnormalized $[p_i, v_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\dim o_i = 11(M-1)+6, \qquad \dim g = 11M. \tag{46}$$

The position actor's six raw self values supply the position and velocity required by its absolute-point interface. This information does not remove the minimum free-flight speed or alter the collision transition. The dimensions describe the native 3D particle; planar compatibility is confined to the setting stated above.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For the particle, collision prediction follows old-velocity line segments. Consequently, the selected acceleration changes subsequent velocity and interception geometry after the current position advance. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

H. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned} h'_k &= h^0_k, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d^{\mathrm{term}}_t &= \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \le 0]. \end{aligned} \tag{47}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma = 1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d^{\mathrm{term}}_t, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{48}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

IX. Theme 5 / Fixed wing / Acceleration / Survival

The Survival defense task couples aerodynamic motion to interception and asset protection. The following wrench and action-to-control map determine the physical transition; its resulting partial information and defender payoff complete the same formulation.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Aerosonde aerodynamic wrench

Let $C_f = \mathrm{diag}(1,-1,-1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \mathrm{atan2}(w, u)$, $\beta = \arcsin \mathrm{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\widehat{p} = bp_b/(2\bar{V})$, $\widehat{q} = cq_b/(2\bar{V})$, $\widehat{r} = br_b/(2\bar{V})$.

$$
\begin{aligned}
E_- &= e^{\mathrm{clip}(-50(\alpha-.47),-500,500)},\\
E_+ &= e^{\mathrm{clip}(50(\alpha+.47),-500,500)},\\
\sigma &= (1+E_-+E_+)/[(1+E_-)(1+E_+)],\\
C_L &= (1-\sigma)(.23+5.61\alpha)\\
&\quad + 2\sigma\,\mathrm{sign}(\alpha)\sin^2\alpha\cos\alpha + 7.95\widehat{q} + .13\delta_e,\\
C_D &= .043 + \frac{(.23+5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e,\\
C_Y &= -.98\beta + .075\delta_a + .19\delta_r,\\
C_\ell &= -.13\beta - .51\widehat{p} + .25\widehat{r} + .17\delta_a + .0024\delta_r,\\
C_m &= .0135 - 2.74\alpha - 38.21\widehat{q} - .99\delta_e,\\
C_n &= .073\beta + .069\widehat{p} - .095\widehat{r} - .011\delta_a - .069\delta_r.
\end{aligned}
\tag{49}
$$

$$
F_B = C_f \left\{ \bar{Q}S \begin{bmatrix} -C_D\cos\alpha + C_L\sin\alpha \\ C_Y \\ -C_D\sin\alpha - C_L\cos\alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \tag{50}
$$

$$
M_B = C_f \left\{ \bar{Q}S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \tag{51}
$$

C. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m^2, $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m^3,

$$
J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \tag{52}
$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145\,2\pi)$, the steady motor root is

$$
\begin{aligned}
A &= \rho D_p^5 C_{Q0}/(4\pi^2),\\
B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m,\\
C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0,\\
\Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.
\end{aligned}
\tag{53}
$$

Set $z = (D_p^4\Omega_p^2/(4\pi^2), D_p^3 V\Omega_p/(2\pi), D_p^2 V^2)$,

$$
\begin{aligned}
T_p &= \rho(.09357, -.06044, -.1079)z,\\
Q_p &= \rho D_p(.005230, .004970, -.01664)z.
\end{aligned}
\tag{54}
$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Action and reference conversion

Choose $a_i \in [-1,1]^3$ or one of the 27 unit primitives. Scale once to $\alpha_i = 5a_i$, then hold $v_d = v_i(t) + \Delta\alpha_i$. At each RK4 stage,

$$
\begin{aligned}
\chi_d &= \mathrm{atan2}(v_{dy}, v_{dx}),\\
\gamma_d &= \mathrm{atan2}(v_{dz}, \max(\|v_{d,xy}\|, 10^{-6})),\\
V_d &= \|v_d\|.
\end{aligned}
\tag{55}
$$

For tiny horizontal reference speed, χ_d is current heading. Component-limited continuous intent need not have norm ≤ 5.

E. Reference tracking and actuator control

Define bank $\phi = \mathrm{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \mathrm{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \mathrm{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \mathrm{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$
\begin{aligned}
V_d &= \mathrm{clip}(V_d, 18, 30),\\
\gamma_d &= \mathrm{clip}(\gamma_d, -15^\circ, 15^\circ),\\
e_\chi &= \mathrm{wrap}(\chi_d - \chi),\\
\bar{\phi} &= -\mathrm{atan2}(\max(V_a, 18)e_\chi, g),\\
\phi_d &= \mathrm{clip}(\bar{\phi}, -45^\circ, 45^\circ),\\
\bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f),\\
\theta_d &= \mathrm{clip}(\bar{\theta}, -20^\circ, 20^\circ),\\
\delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b,\\
\delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b,\\
r_d &= g\tan\phi_d/\max(V_a, 18),\\
\delta_r &= \delta_r^* - .8(\beta - \beta^*) + .1(r_b - r_d),\\
\tau &= \tau^* + .08(V_d - V_a) + .7\sin\gamma_d.
\end{aligned}
\tag{56}
$$

All angles use radians. Starred values are the level trim at V_d, linearly interpolated from exact solutions on a 0.25 m/s grid over $[18, 30]$ m/s; β^* is the trim sideslip. Since $C_{n\delta_r} < 0$, the rudder term opposes sideslip. Divide surfaces by 25°, map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

F. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

G. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{57}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

H. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

I. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{58}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

J. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

K. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{59}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

X. Theme 6 / Fixed wing / Acceleration / Damage

The Damage defense task couples aerodynamic motion to interception and asset protection. The following wrench and action-to-control map determine the physical transition; its resulting partial information and defender payoff complete the same formulation.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Aerosonde aerodynamic wrench

Let $C_f = \mathrm{diag}(1,-1,-1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \mathrm{atan2}(w, u)$, $\beta = \arcsin \mathrm{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\widehat{p} = bp_b/(2\bar{V})$, $\widehat{q} = cq_b/(2\bar{V})$, $\widehat{r} = br_b/(2\bar{V})$.

$$\begin{aligned}
E_- &= e^{\mathrm{clip}(-50(\alpha-.47),-500,500)},\\
E_+ &= e^{\mathrm{clip}(50(\alpha+.47),-500,500)},\\
\sigma &= (1+E_-+E_+)/[(1+E_-)(1+E_+)],\\
C_L &= (1-\sigma)(.23+5.61\alpha)\\
&\quad + 2\sigma\,\mathrm{sign}(\alpha)\sin^2\alpha\cos\alpha + 7.95\widehat{q} + .13\delta_e,\\
C_D &= .043 + \frac{(.23+5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e,\\
C_Y &= -.98\beta + .075\delta_a + .19\delta_r,\\
C_\ell &= -.13\beta - .51\widehat{p} + .25\widehat{r} + .17\delta_a + .0024\delta_r,\\
C_m &= .0135 - 2.74\alpha - 38.21\widehat{q} - .99\delta_e,\\
C_n &= .073\beta + .069\widehat{p} - .095\widehat{r} - .011\delta_a - .069\delta_r.
\end{aligned} \tag{60}$$

$$F_B = C_f \left\{ \bar{Q}S \begin{bmatrix} -C_D\cos\alpha + C_L\sin\alpha \\ C_Y \\ -C_D\sin\alpha - C_L\cos\alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \tag{61}$$

$$M_B = C_f \left\{ \bar{Q}S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \tag{62}$$

C. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m^2, $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m^3,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \tag{63}$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145\,2\pi)$, the steady motor root is

$$\begin{aligned}
A &= \rho D_p^5 C_{Q0}/(4\pi^2),\\
B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m,\\
C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0,\\
\Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.
\end{aligned} \tag{64}$$

Set $z = (D_p^4\Omega_p^2/(4\pi^2), D_p^3 V\Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned}
T_p &= \rho(.09357, -.06044, -.1079)z,\\
Q_p &= \rho D_p(.005230, .004970, -.01664)z.
\end{aligned} \tag{65}$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Action and reference conversion

Choose $a_i \in [-1,1]^3$ or one of the 27 unit primitives. Scale once to $\alpha_i = 5a_i$, then hold $v_d = v_i(t) + \Delta\alpha_i$. At each RK4 stage,

$$\begin{aligned}
\chi_d &= \mathrm{atan2}(v_{dy}, v_{dx}),\\
\gamma_d &= \mathrm{atan2}(v_{dz}, \max(\|v_{d,xy}\|, 10^{-6})),\\
V_d &= \|v_d\|.
\end{aligned} \tag{66}$$

For tiny horizontal reference speed, χ_d is current heading. Component-limited continuous intent need not have norm ≤ 5.

E. Reference tracking and actuator control

Define bank $\phi = \mathrm{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \mathrm{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \mathrm{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \mathrm{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned}
V_d &= \mathrm{clip}(V_d, 18, 30),\\
\gamma_d &= \mathrm{clip}(\gamma_d, -15^\circ, 15^\circ),\\
e_\chi &= \mathrm{wrap}(\chi_d - \chi),\\
\bar{\phi} &= -\mathrm{atan2}(\max(V_a, 18)e_\chi, g),\\
\phi_d &= \mathrm{clip}(\bar{\phi}, -45^\circ, 45^\circ),\\
\bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f),\\
\theta_d &= \mathrm{clip}(\bar{\theta}, -20^\circ, 20^\circ),\\
\delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b,\\
\delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b,\\
r_d &= g\tan\phi_d/\max(V_a, 18),\\
\delta_r &= \delta_r^* - .8(\beta - \beta^*) + .1(r_b - r_d),\\
\tau &= \tau^* + .08(V_d - V_a) + .7\sin\gamma_d.
\end{aligned} \tag{67}$$

All angles use radians. Starred values are the level trim at V_d, linearly interpolated from exact solutions on a 0.25 m/s grid over $[18, 30]$ m/s; β^* is the trim sideslip. Since $C_{n\delta_r} < 0$, the rudder term opposes sideslip. Divide surfaces by 25°, map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

F. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

G. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned}\dim o_i &= 11(M-1)+13,\\ \dim g &= 11M+13N+3.\end{aligned} \tag{68}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t}=\mathbf{1}[h_{i,t}>0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

H. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f=r_0$ by default, with configurable $0<r_f\le r_0$, and a source may damage several enemies.

I. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned}&h'_k=h^0_k, && D'_k=D_k+L_k,\\ &R_t=-\sum_k L_k, && d^{\mathrm{term}}_t=\mathbf{1}[\forall b\in\mathcal{B}_A: h'_b\le 0].\end{aligned} \tag{69}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

J. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i\mid\tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma=1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

K. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned}b_t&=1-d^{\mathrm{term}}_t,\\ y_t&=R_t+\gamma b_t Q^-(\xi_{t+1},a'_R).\end{aligned} \tag{70}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XI. Theme 7 / Fixed wing / Actuator / Survival

The Survival defense task couples aerodynamic motion to interception and asset protection. The following wrench and action-to-control map determine the physical transition; its resulting partial information and defender payoff complete the same formulation.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Aerosonde aerodynamic wrench

Let $C_f = \mathrm{diag}(1,-1,-1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \mathrm{atan2}(w, u)$, $\beta = \arcsin \mathrm{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\widehat{p} = b p_b/(2\bar{V})$, $\widehat{q} = c q_b/(2\bar{V})$, $\widehat{r} = b r_b/(2\bar{V})$.

$$
\begin{aligned}
E_- &= e^{\mathrm{clip}(-50(\alpha-.47),-500,500)},\\
E_+ &= e^{\mathrm{clip}(50(\alpha+.47),-500,500)},\\
\sigma &= (1+E_-+E_+)/[(1+E_-)(1+E_+)],\\
C_L &= (1-\sigma)(.23+5.61\alpha)\\
&\quad + 2\sigma\,\mathrm{sign}(\alpha)\sin^2\alpha\cos\alpha + 7.95\widehat{q} + .13\delta_e,\\
C_D &= .043 + \frac{(.23+5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e,\\
C_Y &= -.98\beta + .075\delta_a + .19\delta_r,\\
C_\ell &= -.13\beta - .51\widehat{p} + .25\widehat{r} + .17\delta_a + .0024\delta_r,\\
C_m &= .0135 - 2.74\alpha - 38.21\widehat{q} - .99\delta_e,\\
C_n &= .073\beta + .069\widehat{p} - .095\widehat{r} - .011\delta_a - .069\delta_r.
\end{aligned}
\tag{71}
$$

$$
F_B = C_f \left\{ \bar{Q}S \begin{bmatrix} -C_D\cos\alpha + C_L\sin\alpha \\ C_Y \\ -C_D\sin\alpha - C_L\cos\alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \tag{72}
$$

$$
M_B = C_f \left\{ \bar{Q}S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \tag{73}
$$

C. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m^2, $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m^3,

$$
J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \tag{74}
$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145\,2\pi)$, the steady motor root is

$$
\begin{aligned}
A &= \rho D_p^5 C_{Q0}/(4\pi^2),\\
B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m,\\
C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0,\\
\Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.
\end{aligned}
\tag{75}
$$

Set $z = (D_p^4\Omega_p^2/(4\pi^2), D_p^3 V\Omega_p/(2\pi), D_p^2 V^2)$,

$$
\begin{aligned}
T_p &= \rho(.09357, -.06044, -.1079)z,\\
Q_p &= \rho D_p(.005230, .004970, -.01664)z.
\end{aligned}
\tag{76}
$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Direct actuator action

Every living agent selects $a_i \in [-1,1]^4$, ordered elevator, aileron, rudder, throttle:

$$
(\delta_e, \delta_a, \delta_r) = 25^\circ(a_1, a_2, a_3), \qquad \tau = (a_4+1)/2. \tag{77}
$$

The actuator vector is constant for the external step and feeds the aerodynamic/motor equations directly. There is no automatic stabilization or navigation controller. In particular $a = 0$ is neutral surfaces with half throttle, not solved trim. Automatic fire still uses the same separate task rules. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.

E. Geometry and initialization

Default $\lambda = .4$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m. Targets use the scaled region $[-920, -760] \times [-480, 480] \times [200, 600]$ m. Red uses an area-uniform XY annulus $[320, 880]$ m around a sampled target; Blue uses $x \in [0, 1000]$ and full scene y; agent altitude lies in $[80, 600]$ m. UAV openings reject distances $\le \max(30\lambda, d_c + 10^{-6})$. Both teams reset at solved 30 m/s level flight; Blue heads west, Red east under random initialization. Only 3D is supported.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Actuator controls do not add an observation block. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{78}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

H. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), \qquad & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{79}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{80}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XII. Theme 8 / Fixed wing / Actuator / Damage

The Damage defense task couples aerodynamic motion to interception and asset protection. The following wrench and action-to-control map determine the physical transition; its resulting partial information and defender payoff complete the same formulation.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Aerosonde aerodynamic wrench

Let $C_f = \mathrm{diag}(1,-1,-1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \mathrm{atan2}(w, u)$, $\beta = \arcsin \mathrm{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\widehat{p} = b p_b/(2\bar{V})$, $\widehat{q} = c q_b/(2\bar{V})$, $\widehat{r} = b r_b/(2\bar{V})$.

$$
\begin{aligned}
E_- &= e^{\mathrm{clip}(-50(\alpha-.47),-500,500)},\\
E_+ &= e^{\mathrm{clip}(50(\alpha+.47),-500,500)},\\
\sigma &= (1+E_-+E_+)/[(1+E_-)(1+E_+)],\\
C_L &= (1-\sigma)(.23+5.61\alpha)\\
&\quad + 2\sigma\,\mathrm{sign}(\alpha)\sin^2\alpha\cos\alpha + 7.95\widehat{q} + .13\delta_e,\\
C_D &= .043 + \frac{(.23+5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e,\\
C_Y &= -.98\beta + .075\delta_a + .19\delta_r,\\
C_\ell &= -.13\beta - .51\widehat{p} + .25\widehat{r} + .17\delta_a + .0024\delta_r,\\
C_m &= .0135 - 2.74\alpha - 38.21\widehat{q} - .99\delta_e,\\
C_n &= .073\beta + .069\widehat{p} - .095\widehat{r} - .011\delta_a - .069\delta_r.
\end{aligned}
\tag{81}
$$

$$
F_B = C_f \left\{ \bar{Q}S \begin{bmatrix} -C_D\cos\alpha + C_L\sin\alpha \\ C_Y \\ -C_D\sin\alpha - C_L\cos\alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \tag{82}
$$

$$
M_B = C_f \left\{ \bar{Q}S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \tag{83}
$$

C. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m^2, $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m^3,

$$
J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \tag{84}
$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145\,2\pi)$, the steady motor root is

$$
\begin{aligned}
A &= \rho D_p^5 C_{Q0}/(4\pi^2),\\
B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m,\\
C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0,\\
\Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.
\end{aligned}
\tag{85}
$$

Set $z = (D_p^4\Omega_p^2/(4\pi^2), D_p^3 V\Omega_p/(2\pi), D_p^2 V^2)$,

$$
\begin{aligned}
T_p &= \rho(.09357, -.06044, -.1079)z,\\
Q_p &= \rho D_p(.005230, .004970, -.01664)z.
\end{aligned}
\tag{86}
$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Direct actuator action

Every living agent selects $a_i \in [-1,1]^4$, ordered elevator, aileron, rudder, throttle:

$$
(\delta_e, \delta_a, \delta_r) = 25^\circ(a_1, a_2, a_3), \qquad \tau = (a_4+1)/2. \tag{87}
$$

The actuator vector is constant for the external step and feeds the aerodynamic/motor equations directly. There is no automatic stabilization or navigation controller. In particular $a = 0$ is neutral surfaces with half throttle, not solved trim. Automatic fire still uses the same separate task rules. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.

E. Geometry and initialization

Default $\lambda = .4$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m. Targets use the scaled region $[-920, -760] \times [-480, 480] \times [200, 600]$ m. Red uses an area-uniform XY annulus $[320, 880]$ m around a sampled target; Blue uses $x \in [0, 1000]$ and full scene y; agent altitude lies in $[80, 600]$ m. UAV openings reject distances $\le \max(30\lambda, d_c + 10^{-6})$. Both teams reset at solved 30 m/s level flight; Blue heads west, Red east under random initialization. Only 3D is supported.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned}\dim o_i &= 11(M-1)+13,\\ \dim g &= 11M+13N+3.\end{aligned} \tag{88}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t}=\mathbf{1}[h_{i,t}>0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f=r_0$ by default, with configurable $0<r_f\le r_0$, and a source may damage several enemies.

H. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned}&h'_k=h^0_k, \qquad D'_k=D_k+L_k,\\ &R_t=-\sum_k L_k, \quad d_t^{\mathrm{term}}=\mathbf{1}[\forall b\in\mathcal{B}_A: h'_b\le 0].\end{aligned} \tag{89}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i\mid\tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma=1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned}b_t &= 1-d_t^{\mathrm{term}},\\ y_t &= R_t+\gamma b_t Q^-(\xi_{t+1}, a'_R).\end{aligned} \tag{90}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XIII. Theme 9 / Fixed wing / Position / Survival

The Survival defense task couples aerodynamic motion to interception and asset protection. The following wrench and action-to-control map determine the physical transition; its resulting partial information and defender payoff complete the same formulation.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Aerosonde aerodynamic wrench

Let $C_f = \mathrm{diag}(1,-1,-1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \mathrm{atan2}(w, u)$, $\beta = \arcsin \mathrm{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\widehat{p} = bp_b/(2\bar{V})$, $\widehat{q} = cq_b/(2\bar{V})$, $\widehat{r} = br_b/(2\bar{V})$.

$$\begin{aligned}
E_- &= e^{\mathrm{clip}(-50(\alpha-.47),-500,500)},\\
E_+ &= e^{\mathrm{clip}(50(\alpha+.47),-500,500)},\\
\sigma &= (1+E_-+E_+)/[(1+E_-)(1+E_+)],\\
C_L &= (1-\sigma)(.23+5.61\alpha)\\
&\quad + 2\sigma\,\mathrm{sign}(\alpha)\sin^2\alpha\cos\alpha + 7.95\widehat{q} + .13\delta_e,\\
C_D &= .043 + \frac{(.23+5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e,\\
C_Y &= -.98\beta + .075\delta_a + .19\delta_r,\\
C_\ell &= -.13\beta - .51\widehat{p} + .25\widehat{r} + .17\delta_a + .0024\delta_r,\\
C_m &= .0135 - 2.74\alpha - 38.21\widehat{q} - .99\delta_e,\\
C_n &= .073\beta + .069\widehat{p} - .095\widehat{r} - .011\delta_a - .069\delta_r.
\end{aligned} \tag{91}$$

$$F_B = C_f \left\{ \bar{Q}S \begin{bmatrix} -C_D\cos\alpha + C_L\sin\alpha \\ C_Y \\ -C_D\sin\alpha - C_L\cos\alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \tag{92}$$

$$M_B = C_f \left\{ \bar{Q}S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \tag{93}$$

C. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m^2, $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m^3,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \tag{94}$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145\,2\pi)$, the steady motor root is

$$\begin{aligned}
A &= \rho D_p^5 C_{Q0}/(4\pi^2),\\
B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m,\\
C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0,\\
\Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.
\end{aligned} \tag{95}$$

Set $z = (D_p^4\Omega_p^2/(4\pi^2), D_p^3 V\Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned}
T_p &= \rho(.09357, -.06044, -.1079)z,\\
Q_p &= \rho D_p(.005230, .004970, -.01664)z.
\end{aligned} \tag{96}$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Absolute position action

Choose $a_i = p_d \in \mathcal{W}$ in metres. At every stage, $d = p_d - p$ determines

$$\begin{aligned}
\chi_d &= \mathrm{atan2}(d_y, d_x),\\
\gamma_d &= \mathrm{atan2}(d_z, \max(\|d_{xy}\|, 30)), \quad V_d = 30.
\end{aligned} \tag{97}$$

Tiny horizontal displacement uses current heading. The aircraft flies toward and through the target; it cannot hover there.

E. Reference tracking and actuator control

Define bank $\phi = \mathrm{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \mathrm{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \mathrm{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \mathrm{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned}
V_d &= \mathrm{clip}(V_d, 18, 30),\\
\gamma_d &= \mathrm{clip}(\gamma_d, -15^\circ, 15^\circ),\\
e_\chi &= \mathrm{wrap}(\chi_d - \chi),\\
\bar{\phi} &= -\mathrm{atan2}(\max(V_a, 18)e_\chi, g),\\
\phi_d &= \mathrm{clip}(\bar{\phi}, -45^\circ, 45^\circ),\\
\bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f),\\
\theta_d &= \mathrm{clip}(\bar{\theta}, -20^\circ, 20^\circ),\\
\delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b,\\
\delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b,\\
r_d &= g\tan\phi_d/\max(V_a, 18),\\
\delta_r &= \delta_r^* - .8(\beta - \beta^*) + .1(r_b - r_d),\\
\tau &= \tau^* + .08(V_d - V_a) + .7\sin\gamma_d.
\end{aligned} \tag{98}$$

All angles use radians. Starred values are the level trim at V_d, linearly interpolated from exact solutions on a 0.25 m/s grid over $[18, 30]$ m/s; β^* is the trim sideslip. Since $C_{n\delta_r} < 0$, the rudder term opposes sideslip. Divide surfaces by 25°, map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

F. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

G. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{99}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

H. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

I. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10 z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{100}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

J. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

K. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{101}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XIV. Theme 10 / Fixed wing / Position / Damage

The Damage defense task couples aerodynamic motion to interception and asset protection. The following wrench and action-to-control map determine the physical transition; its resulting partial information and defender payoff complete the same formulation.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Aerosonde aerodynamic wrench

Let $C_f = \mathrm{diag}(1,-1,-1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \mathrm{atan2}(w, u)$, $\beta = \arcsin \mathrm{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\widehat{p} = bp_b/(2\bar{V})$, $\widehat{q} = cq_b/(2\bar{V})$, $\widehat{r} = br_b/(2\bar{V})$.

$$
\begin{aligned}
E_- &= e^{\mathrm{clip}(-50(\alpha-.47),-500,500)},\\
E_+ &= e^{\mathrm{clip}(50(\alpha+.47),-500,500)},\\
\sigma &= (1+E_-+E_+)/[(1+E_-)(1+E_+)],\\
C_L &= (1-\sigma)(.23+5.61\alpha)\\
&\quad + 2\sigma\,\mathrm{sign}(\alpha)\sin^2\alpha\cos\alpha + 7.95\widehat{q} + .13\delta_e,\\
C_D &= .043 + \frac{(.23+5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e,\\
C_Y &= -.98\beta + .075\delta_a + .19\delta_r,\\
C_\ell &= -.13\beta - .51\widehat{p} + .25\widehat{r} + .17\delta_a + .0024\delta_r,\\
C_m &= .0135 - 2.74\alpha - 38.21\widehat{q} - .99\delta_e,\\
C_n &= .073\beta + .069\widehat{p} - .095\widehat{r} - .011\delta_a - .069\delta_r.
\end{aligned}
\tag{102}
$$

$$
F_B = C_f \left\{ \bar{Q}S \begin{bmatrix} -C_D\cos\alpha + C_L\sin\alpha \\ C_Y \\ -C_D\sin\alpha - C_L\cos\alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \tag{103}
$$

$$
M_B = C_f \left\{ \bar{Q}S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \tag{104}
$$

C. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m^2, $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m^3,

$$
J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \tag{105}
$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145\,2\pi)$, the steady motor root is

$$
\begin{aligned}
A &= \rho D_p^5 C_{Q0}/(4\pi^2),\\
B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m,\\
C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0,\\
\Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.
\end{aligned}
\tag{106}
$$

Set $z = (D_p^4\Omega_p^2/(4\pi^2), D_p^3 V\Omega_p/(2\pi), D_p^2 V^2)$,

$$
\begin{aligned}
T_p &= \rho(.09357, -.06044, -.1079)z,\\
Q_p &= \rho D_p(.005230, .004970, -.01664)z.
\end{aligned}
\tag{107}
$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Absolute position action

Choose $a_i = p_d \in \mathcal{W}$ in metres. At every stage, $d = p_d - p$ determines

$$
\begin{aligned}
\chi_d &= \mathrm{atan2}(d_y, d_x),\\
\gamma_d &= \mathrm{atan2}(d_z, \max(\|d_{xy}\|, 30)), \quad V_d = 30.
\end{aligned}
\tag{108}
$$

Tiny horizontal displacement uses current heading. The aircraft flies toward and through the target; it cannot hover there.

E. Reference tracking and actuator control

Define bank $\phi = \mathrm{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \mathrm{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \mathrm{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \mathrm{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$
\begin{aligned}
V_d &= \mathrm{clip}(V_d, 18, 30),\\
\gamma_d &= \mathrm{clip}(\gamma_d, -15^\circ, 15^\circ),\\
e_\chi &= \mathrm{wrap}(\chi_d - \chi),\\
\bar{\phi} &= -\mathrm{atan2}(\max(V_a, 18)e_\chi, g),\\
\phi_d &= \mathrm{clip}(\bar{\phi}, -45^\circ, 45^\circ),\\
\bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f),\\
\theta_d &= \mathrm{clip}(\bar{\theta}, -20^\circ, 20^\circ),\\
\delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b,\\
\delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b,\\
r_d &= g\tan\phi_d/\max(V_a, 18),\\
\delta_r &= \delta_r^* - .8(\beta - \beta^*) + .1(r_b - r_d),\\
\tau &= \tau^* + .08(V_d - V_a) + .7\sin\gamma_d.
\end{aligned}
\tag{109}
$$

All angles use radians. Starred values are the level trim at V_d, linearly interpolated from exact solutions on a 0.25 m/s grid over $[18, 30]$ m/s; β^* is the trim sideslip. Since $C_{n\delta_r} < 0$, the rudder term opposes sideslip. Divide surfaces by 25°, map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

F. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

G. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{110}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t}=\mathbf{1}[h_{i,t}>0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

H. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f=r_0$ by default, with configurable $0<r_f\le r_0$, and a source may damage several enemies.

I. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned} &h'_k=h^0_k, \qquad\qquad D'_k=D_k+L_k, \\ &R_t=-\sum_k L_k, \quad d_t^{\mathrm{term}}=\mathbf{1}[\forall b\in\mathcal{B}_A: h'_b\le 0]. \end{aligned} \tag{111}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

J. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i\mid\tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma=1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

K. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1-d_t^{\mathrm{term}}, \\ y_t &= R_t+\gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{112}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XV. Theme 11 / Quadrotor / Acceleration / Survival

For the Survival defense task, rotor thrust and attitude determine how navigation changes the interception geometry. We specify the controlled rigid-body transition and initial law before connecting this motion to the defender information and task objective.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0,1]$. With $m = 4.34$ kg, $J = \mathrm{diag}(.0820, .0845, .1377)$ kg m^2, arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \qquad f_j = f_{\max} u_j, \tag{113}$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \tag{114}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \tag{115}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z). Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

C. Acceleration intent

Each living agent chooses either $a_i \in [-1,1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1,0,1\}^3 \setminus \{0\}\}, \tag{116}$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 6a_i$ m/s^2. Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}\,6$; discrete nonzero intents have norm 6. 6 is a control scale, not a clamp on actual rigid-body acceleration. The external intent establishes $v_d = v(t) + 6\Delta a_i$ once. This reference is held for the full external step. It is not recomputed as $v_{\text{stage}} + 6\Delta a_i$ at each stage. The controller below maps the reference to feasible rotor commands; actual acceleration depends on the evolving attitude and rotor limits.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \mathrm{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \tfrac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81 e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \tag{117}$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s^2 bound is below g, so F_d and its heading construction are nonsingular. The attitude gains 8.81 and 2.54 are source-derived [2]; this feedback uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \mathrm{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \qquad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \tag{118}$$

$$u_j = \mathrm{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right). \tag{119}$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.

E. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \le r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, rejecting draws outside the scene; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{120}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

H. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{121}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{122}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XVI. Theme 12 / Quadrotor / Acceleration / Damage

For the Damage defense task, rotor thrust and attitude determine how navigation changes the interception geometry. We specify the controlled rigid-body transition and initial law before connecting this motion to the defender information and task objective.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0,1]$. With $m = 4.34$ kg, $J = \mathrm{diag}(.0820, .0845, .1377)$ kg m^2, arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \qquad f_j = f_{\max} u_j, \tag{123}$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \tag{124}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \tag{125}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z). Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

C. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \tag{126}$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 6a_i$ m/s^2. Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}\,6$; discrete nonzero intents have norm 6. 6 is a control scale, not a clamp on actual rigid-body acceleration. The external intent establishes $v_d = v(t) + 6\Delta a_i$ once. This reference is held for the full external step. It is not recomputed as $v_{\text{stage}} + 6\Delta a_i$ at each stage. The controller below maps the reference to feasible rotor commands; actual acceleration depends on the evolving attitude and rotor limits.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \mathrm{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \tfrac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81 e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \tag{127}$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s^2 bound is below g, so F_d and its heading construction are nonsingular. The attitude gains 8.81 and 2.54 are source-derived [2]; this feedback uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \mathrm{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \qquad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \tag{128}$$

$$u_j = \mathrm{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right). \tag{129}$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.

E. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \le r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, rejecting draws outside the scene; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned}\dim o_i &= 11(M-1)+13,\\ \dim g &= 11M+13N+3.\end{aligned} \tag{130}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t}=\mathbf{1}[h_{i,t}>0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f=r_0$ by default, with configurable $0<r_f\le r_0$, and a source may damage several enemies.

H. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned}&h'_k=h^0_k, && D'_k=D_k+L_k,\\ &R_t=-\sum_k L_k, && d_t^{\mathrm{term}}=\mathbf{1}[\forall b\in\mathcal{B}_A: h'_b\le 0].\end{aligned} \tag{131}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i\mid\tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma=1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned}b_t&=1-d_t^{\mathrm{term}},\\ y_t&=R_t+\gamma b_t Q^-(\xi_{t+1},a'_R).\end{aligned} \tag{132}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XVII. Theme 13 / Quadrotor / Actuator / Survival

For the Survival defense task, rotor thrust and attitude determine how navigation changes the interception geometry. We specify the controlled rigid-body transition and initial law before connecting this motion to the defender information and task objective.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \mathrm{diag}(.0820, .0845, .1377)$ kg m^2, arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \qquad f_j = f_{\max} u_j, \tag{133}$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \tag{134}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \tag{135}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z). Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

C. Direct rotor-thrust action

Each living agent selects $a_i \in [0, 1]^4$, ordered front, right, rear, left, and $u = a_i$ is held throughout the external step. The wrench map enters Newton-Euler dynamics directly, without automatic attitude or position stabilization. Zero thrust is free fall, not hover; .4 on every rotor is ideal level hover. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.The resulting control burden is rotor-level feedback from local information. Identical rotor actions can generate different world accelerations because thrust follows the current body $+z$ axis. Attitude and angular velocity are therefore indispensable components of the physical state and are explicitly supplied in the actor's self block.

A saturated or destabilizing action is propagated by the rigid-body equations. Low altitude does not itself cause a crash: scene boundaries project position and remove outward velocity. Finite asset health follows the task-specific evolution defined below.

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \le r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, rejecting draws outside the scene; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Rotor controls do not add an observation block. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{136}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

F. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

G. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{137}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

H. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

I. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{138}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XVIII. Theme 14 / Quadrotor / Actuator / Damage

For the Damage defense task, rotor thrust and attitude determine how navigation changes the interception geometry. We specify the controlled rigid-body transition and initial law before connecting this motion to the defender information and task objective.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \mathrm{diag}(.0820, .0845, .1377)$ kg m^2, arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \qquad f_j = f_{\max} u_j, \tag{139}$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \tag{140}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \tag{141}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z). Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

C. Direct rotor-thrust action

Each living agent selects $a_i \in [0, 1]^4$, ordered front, right, rear, left, and $u = a_i$ is held throughout the external step. The wrench map enters Newton-Euler dynamics directly, without automatic attitude or position stabilization. Zero thrust is free fall, not hover; .4 on every rotor is ideal level hover. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.The resulting control burden is rotor-level feedback from local information. Identical rotor actions can generate different world accelerations because thrust follows the current body $+z$ axis. Attitude and angular velocity are therefore indispensable components of the physical state and are explicitly supplied in the actor's self block.

A saturated or destabilizing action is propagated by the rigid-body equations. Low altitude does not itself cause a crash: scene boundaries project position and remove outward velocity. Finite asset health follows the task-specific evolution defined below.

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, rejecting draws outside the scene; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{142}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t}=\mathbf{1}[h_{i,t}>0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

F. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f=r_0$ by default, with configurable $0<r_f\le r_0$, and a source may damage several enemies.

G. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned} &h'_k=h^0_k, \qquad\qquad D'_k=D_k+L_k, \\ &R_t=-\sum_k L_k, \quad d_t^{\mathrm{term}}=\mathbf{1}[\forall b\in\mathcal{B}_A: h'_b\le 0]. \end{aligned} \tag{143}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

H. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i\mid\tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma=1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

I. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1-d_t^{\mathrm{term}}, \\ y_t &= R_t+\gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{144}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XIX. Theme 15 / Quadrotor / Position / Survival

For the Survival defense task, rotor thrust and attitude determine how navigation changes the interception geometry. We specify the controlled rigid-body transition and initial law before connecting this motion to the defender information and task objective.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0,1]$. With $m = 4.34$ kg, $J = \mathrm{diag}(.0820, .0845, .1377)$ kg m^2, arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \qquad f_j = f_{\max} u_j, \tag{145}$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \tag{146}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \tag{147}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z). Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

C. Absolute position intent

Each living agent selects $p_d \in \mathcal{W}$ in metres. At every RK4 stage compute

$$\begin{aligned} v_d &= \mathrm{sat}_{12}(.5(p_d - p)), \\ \mathrm{sat}_L(z) &= z \min(1, L/\|z\|). \end{aligned} \tag{148}$$

For $z = 0$, define $\mathrm{sat}_L(0) = 0$. The position reference is held over the external step; feedback updates using stage position and velocity. Unlike the fixed wing and minimum-speed particle, this ideal quadrotor can command zero velocity near its target. Reaching the point is a feedback outcome, not an instantaneous assignment.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \mathrm{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \tfrac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81 e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \tag{149}$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s^2 bound is below g, so F_d and its heading construction are nonsingular. The attitude gains 8.81 and 2.54 are source-derived [2]; this feedback uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \mathrm{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \qquad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \tag{150}$$

$$u_j = \mathrm{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1 \right). \tag{151}$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.

E. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \le r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, rejecting draws outside the scene; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{152}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f = r_0$ by default, with configurable $0 < r_f \le r_0$, and a source may damage several enemies.

H. Survival payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Survival task defines the shared defender reward and world completion. The Survival outcome $z(s')$ is (14): asset loss at $h'_k < 10^{-3}$ has priority over elimination of all Blue Attack agents at $h'_b < 10^{-3}$.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10 z(s'), & d_t^{\mathrm{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \tag{153}$$

Blue receives $-R_t$. Red elimination alone does not complete the world. An actor with nonpositive health retires after its lethal final transition, while the team return continues until natural world completion. The optional weights in Section IV replace this native scalar; they do not add a second terminal bonus.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i \mid \tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Survival, Red maximizes the discounted terminal asset-protection payoff; at $\gamma = 1$, successful defense contributes $+10$ and loss of any asset contributes -10. Interception is valuable through this team outcome rather than an additional native flight reward. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1 - d_t^{\mathrm{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{154}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.

XX. Theme 16 / Quadrotor / Position / Damage

For the Damage defense task, rotor thrust and attitude determine how navigation changes the interception geometry. We specify the controlled rigid-body transition and initial law before connecting this motion to the defender information and task objective.

A. Physical model and controlled transition

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state (1) includes this vehicle state, all other entities, health, roles, side, the clock, task variables and relevant memory. The controlled motion follows Newton–Euler (2) and RK4 (4), with the wrench and action-to-control map specified below. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

B. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \mathrm{diag}(.0820, .0845, .1377)$ kg m^2, arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \qquad f_j = f_{\max} u_j, \tag{155}$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \tag{156}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \tag{157}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z). Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

C. Absolute position intent

Each living agent selects $p_d \in \mathcal{W}$ in metres. At every RK4 stage compute

$$\begin{aligned} v_d &= \mathrm{sat}_{12}(.5(p_d - p)), \\ \mathrm{sat}_L(z) &= z \min(1, L/\|z\|). \end{aligned} \tag{158}$$

For $z = 0$, define $\mathrm{sat}_L(0) = 0$. The position reference is held over the external step; feedback updates using stage position and velocity. Unlike the fixed wing and minimum-speed particle, this ideal quadrotor can command zero velocity near its target. Reaching the point is a feedback outcome, not an instantaneous assignment.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \mathrm{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \tfrac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81 e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \tag{159}$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s^2 bound is below g, so F_d and its heading construction are nonsingular. The attitude gains 8.81 and 2.54 are source-derived [2]; this feedback uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \mathrm{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \qquad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \tag{160}$$

$$u_j = \mathrm{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1 \right). \tag{161}$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it. This action-to-wrench map closes the synchronized 100-substep RK4 transition (4). The external command is held for 1 s; intent feedback is recomputed at each stage, whereas direct actuators remain constant. The boundary rule of Section II clips predicted paths and endpoint positions, and removes outward endpoint velocity.

E. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \le r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, rejecting draws outside the scene; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

F. Defender information

The navigation transition above determines both future interception opportunities and the motion encoded in each defender's observation. The local input is the deterministic encoding (11), with the following interface-specific self information. The actor receives the $(M-1)\times 11$ entity table ($V_{\max}=20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows contain eleven normalized values in fixed roster order; dead rows are zero, and own health is absent from the self block.

$$\begin{aligned} \dim o_i &= 11(M-1)+13, \\ \dim g &= 11M+13N+3. \end{aligned} \tag{162}$$

The raw self block exposes the defender's quaternion and body rate, which are needed to interpret body-frame actuation. Other UAV attitudes and rates remain absent from entity rows even though they affect subsequent rigid-body motion.

Every living entity appears irrespective of firing range or Attack, Scout and Disturb sectors. Entity validity identifies living rows; it differs from the pre-transition action participation mask $\ell_{i,t}=\mathbf{1}[h_{i,t}>0]$. Side flags do not encode functional role, and the encoding omits policy memory and absolute time. Thus noiseless observation does not make the defender's local information Markov.

The privileged feature g follows (12). Its availability is an explicit centralized-training choice; the complete transition state remains (1). In particular, exact value conditioning may require task state and policy history in addition to the packed features.

G. Frozen-opponent transition

With the physical and information maps fixed, the defender process is the Dec-POMDP (13). Its per-step map F_C combines this section's controlled motion with the ordered collision and interaction rules of Section II. The separately specified reset law ρ_0 supplies the initial distribution. Episode-fixed Attack, Scout and Disturb roles are retained. A specified frozen external π_B supplies each living Blue agent's navigation action and updates any Blue memory. Automatic firing is a separate environment rule for both teams; it does not determine Blue navigation.

For this rigid model, collision prediction uses synchronized, clipped RK4 path segments generated by the same action-to-wrench map as the physical transition. Collision victims do not move; combat victims complete the current motion, with inactive velocity zeroed at the next update. Losses use the common post-collision, pre-motion snapshot. The firing threshold is $r_f=r_0$ by default, with configurable $0<r_f\le r_0$, and a source may damage several enemies.

H. Damage payoff and natural completion

This transition produces raw asset loss L_k and successor UAV health, from which the Damage task defines the shared defender reward and world completion. In Damage, assets retain finite reset health. Raw simultaneous contributions are accumulated before health clipping, including overkill, without team-size, asset-count or HP normalization.

$$\begin{aligned} &h'_k=h^0_k, \qquad\qquad D'_k=D_k+L_k, \\ &R_t=-\sum_k L_k, \quad d_t^{\mathrm{term}}=\mathbf{1}[\forall b\in\mathcal{B}_A: h'_b\le 0]. \end{aligned} \tag{163}$$

Blue receives $-R_t$. Each slot receives the full team reward, including dead Red slots and later damage after Red extinction; duplicating it across teammates does not create additional team return. Dead slots have zero observations and ignored actions until global completion or truncation. No Survival terminal bonus or optional Survival weights apply.

I. Decentralized defender objective

A Red policy factors into actors $\pi_i(a_i\mid\tau_i)$ using only their local histories and the action interface defined above. The shared objective of Section IV averages over this reset law, the frozen-opponent transition and Red policy decisions. For Damage, Red maximizes negative discounted raw asset loss; at $\gamma=1$, its return is $-\sum_k D_k$. Navigation and cooperation are therefore evaluated by the accumulated damage they prevent, including any continuation after Red extinction. The frozen Blue policy defines the opposing process throughout this optimization. Joint adaptation of both teams instead gives the corresponding partially observable stochastic game. Optional grouping remains the separate acceleration-execution extension in Section IV.

J. Value conditioning and sampled horizons

Exact centralized targets use ξ_t, which augments $g(s_t)$ with relevant task state and policy history, including Red recurrent memory and frozen Blue memory. Target actions use the corresponding successor policy memories. The shared value construction specializes here to

$$\begin{aligned} b_t &= 1-d_t^{\mathrm{term}}, \\ y_t &= R_t+\gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \tag{164}$$

The horizon H truncates a collected sample without awarding native victory; successor value is retained at truncation and after individual death. Natural completion takes priority if it coincides with H. Entity validity, actor retirement and the global bootstrap are consequently distinct. The action loss uses pre-step participation, so a final lethal action remains trainable. This formulation specifies the defender objective and its value semantics without prescribing an optimizer or claiming a training result.